ELEC70089
ELEC60008

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

CONTROL ENGINEERING

Monday, 12 December 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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CONTROL ENGINEERING

1. Consider the following short questions accounting, overall, for 40% of the exam mark (that is 40 marks out of 100).

a) Let $A \in \mathbb{R}^{n\times n}$ and assume $\det A \neq 0$. Let

$$p(\lambda) = \det(\lambda I - A) = \lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_1\lambda + \alpha_0$$

be its characteristic polynomial. Recall that $\alpha_0 = \det(-A)$.

Using Cayley-Hamilton Theorem determine an expression for A^{-1} in terms of the coefficients α_i of the characteristic polynomial and of powers of A.

[4 marks]

b) Consider the continuous-time linear system described by the equation

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u.$$

Find a piece-wise constant input of the form

$$u(t) = \begin{cases} \alpha, & \text{if } 0 \leq t < 1, \\ \beta, & \text{if } 1 \leq t < 2, \end{cases}$$

with α and β constant, which transfers the initial state $x(0) = [0 \;\; 0]^\top$ to the final state $x(2) = [4 \;\; 0]^\top$.

[6 marks]

Show that it is possible to *define* the input signal for all $t \geq 2$ in such a way that $x(t) = x(2)$, for all $t \geq 2$. [2 marks]

c) Consider the continuous-time nonlinear system described by the equations

$$\dot{x}_1 = -x_1 + x_1x_2, \qquad \dot{x}_2 = -x_2 - x_1x_2 + u, \qquad y = x_1^2 + x_2^2.$$

Let $u = 2$ and determine all equilibrium points of the system. [4 marks]

Write the linearisation of the system around the equilibrium $\bar{x} = [1 \;\; 1]^\top$, that is write explicitly the matrices A, B and C of the linearised system around the given equilibrium. [6 marks]

Using the principle of stability in the first approximation discuss the stability properties of the equilibrium $\bar{x} = [1 \;\; 1]^\top$ of the nonlinear system.

[4 marks]

d) Consider the discrete-time system

$$x_1^+ = x_1 + x_2 + \alpha u, \qquad x_2^+ = -x_1 - x_2 + \beta u, \qquad y = x_1 - 2x_2,$$

with α and β constant.

Show that the system is controllable for all α and β. [2 marks]

Show that the system is observable for all α and β. [2 marks]

Determine a condition on α and β such that the system is reachable. [2 marks]

e) Consider the scalar system (note the non-standard definition of the output variable)

$$x(k+1) = x(k) + u(k) \qquad y(k) = x(k-1),$$

and the system

$$z(k+1) = az(k) + bu(k) \qquad w(k) = z(k) + cy(k).$$

Determine constants a, b, and c such that

$$x(k) = w(k),$$

for all $k \geq K$, independently of the initial conditions $x(-1)$, $x(0)$ and of u, and for some $K > 0$. Determine the minimum value of such a parameter K.
(Hint: consider the signal $e(k) = x(k) - w(k)$.)

[8 marks]

2. Consider the two coupled spring-damper-mass systems in Figure 2.

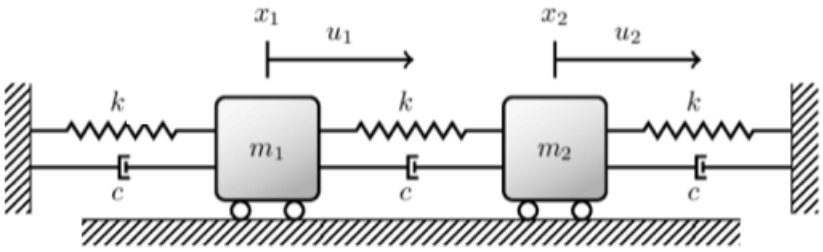


Figure 2: Two coupled spring-damper-mass systems.

This is described by the equations of motion

$$m_1\ddot{x}_1 = -2kx_1 - c\dot{x}_1 + kx_2 + u_1, \qquad m_2\ddot{x}_2 = kx_1 - c\dot{x}_2 - 2kx_2 + u_2,$$

with x_1 and x_2 the position of the masses, m_1 and m_2 the masses of the *blocks*, k the elastic constant of the springs, c the damping coefficient, and u_1 and u_2 external forces.

Assume $m_1 = m_2 = 1$, $c = 0$, and $k > 0$.

a) Let $\xi_1 = x_1$, $\xi_2 = \dot{x}_1$, $\xi_3 = x_2$, $\xi_4 = \dot{x}_2$. Determine a state-space realization of the system, that is write explicitly matrices A and B such that

$$\dot{\xi} = A\xi + Bu,$$

with $\xi = [\xi_1,\ \xi_2,\ \xi_3,\ \xi_4]^\top$ and $u = [u_1,\ u_2]^\top$. [8 marks]

b) Show that the system determined in part a) is controllable. [6 marks]

c) Consider the change of coordinates

$$z_1 = \xi_1 - \xi_3, \quad z_2 = \xi_2 - \xi_4, \quad z_3 = \xi_3, \quad z_4 = \xi_4.$$

Write a state space realization of the system in the z coordinates, that is determine matrices $\tilde{A}$ and $\tilde{B}$ such that

$$\dot{z} = \tilde{A}z + \tilde{B}u.$$

Write explicitly the matrices $\tilde{A}$ and $\tilde{B}$. [6 marks]

(Hint: the matrix $\tilde{A}$ is such that $\tilde{A}_{13} = \tilde{A}_{14} = \tilde{A}_{23} = \tilde{A}_{24} = 0$. There are other zero entries in $\tilde{A}$, but these are *less* important.)

Show that the characteristic polynomial of the matrix $\tilde{A}$ can be factored as the product of two second-order polynomials. Hence show that the system is stable. [4 marks]

d) Assume $u_1 = u_2 = u$. Using the state-space representation determined in part c) show that the resulting system (which is a single input system) is not controllable. Determine the unreachable modes of the system and discuss if the system is stabilizable. [6 marks]

3. Consider a long copper rod, the temperature of which has to be controlled by heating or cooling its endpoints. To simplify the problem, suppose that the temperature profile can be approximated by the temperatures x_1, x_2, and x_3 at three (discrete) points on the rod. The temperatures at the endpoints are the inputs u_1 and u_2. All temperatures are relative to the temperature of the surroundings.

The system is described by the differential equations

$$\begin{aligned} \dot{x}_1 &= \alpha(u_1 - x_1) + \alpha(x_2 - x_1), \\ \dot{x}_2 &= \alpha(x_1 - x_2) + \alpha(x_3 - x_2), \\ \dot{x}_3 &= \alpha(x_2 - x_3) + \alpha(u_2 - x_3), \end{aligned}$$

with $\alpha > 0$ a constant that depends upon the thermal conductivity and the specific heat of the rod.

For simplicity, assume $\alpha = 1$.

a) Consider the problem of controlling the temperatures x_1, x_2, and x_3 along the rod with u_1 only, that is setting $u_2 = 0$.

i) Assume $x(0) = 0$ and let $T > 0$ be given. Is it possible to achieve an arbitrary temperature profile at time T, that is, is it possible to find an input signal $u_1(t)$, for $t \in [0, T]$, such that $x(T) = x^\star$ for any $x^\star$? (Motivate your answer, but do no compute explicitly the input signal.) [4 marks]

ii) Let u_1 be constant. Determine the equilibrium points of the system as a function of u_1. [4 marks]

iii) Parts a.i) and a.ii) seem to contradict each other, that is part a.i) states that any state can be reached from zero, and part a.ii) that only some equilibrium points can be achieved. Explain why this is not a contradiction. [4 marks]

b) Consider the output signal $y = x_2$. Study the observability and detectability properties of the system and determine the unobservable modes. [8 marks]

Design an observer such that the estimation error system has all eigenvalues equal to -2. Explain why this design objective is achievable despite the fact that the system is not observable. [8 marks]

Explain why the output injection matrix which achieves the considered design objective is not unique. [2 marks]